Ahash Thirukumaran

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Education

New Jersey Institute of Technology - *B.S. in Computer Science* May 2026

GPA: 3.3

- **Relevant Coursework:** ML, Computer Architecture, Data Science, Computational Theory, Data Structures & Algorithm, Software Engineering Design, Intensive Programming in Linux

Experience

Software Engineering Intern | *Wakefern Food Corp. Tech Division* Feb 2026 – May 2026

- Building a full-stack **iOS/Android** event app for the ShopRite LPGA Classic with **8 major features** (auth, scheduling, leaderboard, gallery, notifications, sponsor management, interactive map, admin panel) across a dual **Ionic mobile + Angular web** frontend (**41 pages/components**); launching May 15, 2026.
- Architected the backend using **Spring Boot** and containerized the development environment with **Docker** to ensure consistent builds across the team.
- Designed a **13-entity PostgreSQL** schema supporting event scheduling, player 4-some assignments, sponsor tiers, and user profiles.
- Built **57 RESTful API endpoints** serving personalized schedules, tee-time data, real-time scores, and event content to an **Angular + Ionic** frontend.

Computer Science Course Assistant | *NJIT* Jan 2024 – May 2026

- Tutored 100+ students in an Object-Oriented Programming course (**Java**), clarifying complex topics such as **inheritance, polymorphism, and advanced data structures.**
- Held regular office hours and group review sessions, significantly boosting average assignment completion rates.
- Evaluated and graded programming labs, projects, and exams, ensuring consistent application of grading standards.

Projects

Covey.Town UI Feature Development

- Developed a coordinate-based virtual pet guide frontend using animated sprite sheets, real-time movement updates, and world-state synchronization to assist user navigation in Covey.Town's multiplayer environment.
- Collaborated on a 4-person **agile Scrum** team across 4 sprints, participating in sprint planning and stand-ups.
- Deployed via **AWS Lightsail** and maintained **CI/CD** pipelines to automate testing and deployments, reducing deployment errors by **60%** and enabling faster feature delivery.

Language Interpreter

- Built a custom interpreter for a Fortran-like language from scratch in C++, implementing a **recursive-descent parser** to convert source code into an executable program structure.
- Implemented an efficient lexer using strict string tokenization to process source code, handling a robust vocabulary of keywords, relational operators, literals, and identifiers.
- Designed and constructed an **Abstract Syntax Tree (AST)** using C++ class hierarchies and smart pointers to safely represent and manage program semantics without memory leaks.

Secure Sandbox Environment

- Built a lightweight container runtime from scratch in C++ using Linux kernel primitives (clone(), namespaces, chroot, cgroups v2, seccomp), achieving process isolation equivalent to Docker without any container framework.
- Designed and implemented a default-deny **seccomp syscall filter**, reducing the kernel attack surface to ∼100 explicitly whitelisted calls and blocking all filesystem/namespace escape vectors.
- Enforced layered resource limits via **cgroups v2** (50% CPU quota, 256 MB memory, 64 PID cap) with setrlimit as a guaranteed fallback, preventing sandboxed processes from impacting host stability.
- Managed the full privilege lifecycle, dropping root to a dedicated unprivileged user post-setup while selectively preserving CAP_NET_RAW via Linux ambient capabilities to maintain network functionality across execve.

Technical Skills

Languages: Java, Python, C, C++, SQL, JavaScript, HTML, TypeScript
Frameworks: React.js, Node.js, Express.js, Next.js, Spring Boot, Angular, Ionic
Databases: MongoDB, MySQL, PostgreSQL, Redis
Cloud & Tools: AWS, Git, GitHub, Linux/Unix, Docker